

A Transfer Learning-based Method for Probabilistic Forecasting of Residential Load with Deficient Data

Sida Cheng

School of Electrical and Electronic
Engineering

Nanyang Technological University

Singapore

SIDA001@e.ntu.edu.sg

Yan Xu

School of Electrical and Electronic
Engineering

Nanyang Technological University

Singapore

xuyan@ntu.edu.sg

Zhengyuan Du

School of Electrical and Electronic
Engineering

Nanyang Technological University

Singapore

e220188@e.ntu.edu.sg

Abstract—Accurate and uncertainty-aware load forecasting for smart meters is crucial for smart grid operations, given the growing variability introduced by renewable energy and user-side electrification. However, traditional forecasting models often struggle in data-deficient scenarios. They also fail to effectively quantify predictive uncertainty, limiting risk-aware decision-making. This paper presents a transfer learning-based probabilistic forecasting method for data-deficient scenarios. It integrates an LSTM-attention mechanism-based probabilistic forecaster with a domain adaptation process through transfer learning. Initially, smart meter data from a source domain (a data-rich area) is utilized to train an LSTM-attention mechanism-based forecaster for point forecasts and error variances. Subsequently, the pretrained model is fine-tuned with data from the target domain (a data-deficient scenario). Finally, a prediction interval is constructed based on the refined point forecasts and estimated variances by a specialized module. Simulation results demonstrate that the method achieves high forecasting accuracy and effective uncertainty quantification with fewer target domain data compared to benchmarks.

Keywords—Residential load forecasting, probabilistic forecasting, transfer learning, data-deficient.

I. INTRODUCTION

Accurate load forecasting is crucial for the secure and economical operation of modern power systems. The integration of renewable energy and increased electrification have raised uncertainty in electricity consumption. Traditional deterministic methods cannot fully capture this uncertainty, limiting their use in risk-aware planning [1], [2]. Moreover, many forecasting approaches struggle with data scarcity, a common issue in smart meter datasets. Limited data hinder models from effectively capturing diverse, dynamic consumption patterns and generalizing trends, resulting in suboptimal forecasts [3], [4].

In recent years, deep learning techniques have demonstrated superior performance in capturing the nonlinear and temporal dependencies of load data, outperforming classical statistical models such as linear regression and autoregressive methods [5], [6]. Notable deep learning architectures—such as artificial neural networks (ANNs) [5], support vector machines (SVMs) [6], long short-term memory (LSTM) networks [7], and residual networks (ResNet) [8]—have been successfully applied to short-term load forecasting. Building on this foundation, probabilistic forecasting methods leveraging empirical copulas [9], quantile regression with embedded features [10], and deep autoregressive networks [11] have further advanced uncertainty quantification. Nevertheless, a common assumption in these approaches is the availability of large-scale, high-quality historical data—a condition that often does

not hold in practical scenarios such as newly commissioned loads or sparsely monitored areas [12].

To address data scarcity, transfer learning has emerged as a promising technique. It enables the adaptation of models trained on data-rich domains to data-constrained target domains, thereby improving generalization without the need for extensive retraining. Recent advances include meta-learning for few-shot adaptation [13], reinforcement learning for optimization in dynamic environments [14], and domain similarity-based transfer chains [15]. Additionally, multi-layer representation schemes [16] and attention-based frameworks [17] have been proposed to mitigate negative transfer and enhance domain adaptability. Despite these developments, existing research predominantly focuses either on improving probabilistic forecasting under data-rich conditions or enhancing prediction accuracy under data-deficient scenarios—seldom integrating both paradigms to simultaneously address uncertainty and data limitations.

To bridge these two challenges, this paper proposes a data-driven, transfer learning-based probabilistic forecasting method designed specifically for the data-deficient scenarios in smart meter applications. The proposed approach first trains an LSTM-attention mechanism-based probabilistic forecaster on smart meter data from a source domain. This model captures both point forecasts and their associated uncertainties. During deployment, the pretrained model is fine-tuned using smart meter data from the target domain through transfer learning. The final prediction interval is constructed by combining the point forecast and the corresponding estimation variance generated by the model. The contribution of this paper is listed as follows:

- The LSTM-attention mechanism-based probabilistic forecaster enhances accuracy and robustness with limited datasets. The LSTM captures temporal features in smart meter data, while the attention mechanism evaluates time series weights.
- To address the data-deficient issue in practical smart meter load forecasting, a two-stage training framework is introduced. First, the model is trained in a data-rich source domain to learn generalizable probabilistic forecasting strategies. Then, it is fine-tuned using limited datasets from a target domain during deployment via a transfer learning algorithm.

II. PROBLEM FORMULATION

A. Probabilistic Forecasting

In time series forecasting, point forecasting focuses on predicting a single deterministic value by minimizing the prediction error between the forecast and the actual observation. The model is usually expressed as:

$$\hat{y} = f(\mathbf{x}) \quad (1)$$

where $\mathbf{x}$ is the input feature vector, $\hat{y}$ is the predicted value, and $f(\mathbf{x})$ is the prediction function that maps input features to predicted value.

In contrast, probabilistic forecasting seeks not only to provide a predicted value but also the uncertainty of the prediction, typically via a confidence interval:

$$[\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}] = f(\mathbf{x}) \quad (2)$$

where $\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}$ are the lower and upper bounds of the prediction interval.

To simplify this task, this paper assumes that the future load, conditioned on input features, follows a Gaussian distribution: $\mathcal{N}(\mu, \sigma^2)$. This assumption is reasonable since the aggregated load from multiple households tends to follow a near-normal distribution at hourly resolution due to the Central Limit Theorem [18].

The probabilistic forecaster, which estimates both the point prediction value $\hat{y}$ and its corresponding σ^2 error variance, is optimized using a Gaussian-based loss function for uncertainty capture, as detailed below:

$$\hat{y}, \sigma^2 = f_{\text{Gaussian}}(\mathbf{x}) \quad (3)$$

$$\text{IF} = \mathcal{L}_{\text{Gaussian}}(\hat{y}, \sigma^2) \quad (4)$$

where $f_{\text{Gaussian}}(\cdot)$ is the prediction function based on the Gaussian distribution, IF is the uncertainty-aware loss function, and $\mathcal{L}_{\text{Gaussian}}(\cdot)$ is the loss function based on the Gaussian likelihood.

Finally, the probability intervals can be calculated by utilizing the point prediction value and the corresponding estimated error variance.

B. Transfer Learning

In practical scenarios such as load forecasting, traditional deep learning methods often rely on large amounts of data to achieve good predictive performance. Specifically, the model is trained on a large-scale domain dataset D^{large} by minimizing a loss function $\mathcal{L}$ and learns parameters θ of the prediction function:

$$\theta = \arg \min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim D^{\text{large}}} [\mathcal{L}(f(\mathbf{x}), y)] \quad (5)$$

However, in real-world applications, the amount of data available in the target domain is often very limited, which makes it difficult to meet the requirements of data-intensive modeling. To address this issue, this paper adopts a transfer learning algorithm. This method first performs pretraining on a large-scale source domain dataset D_S^{large} to learn general representations:

$$\theta_S = \arg \min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim D_S^{\text{large}}} [\mathcal{L}_S(f(\mathbf{x}), y)] \quad (6)$$

Then, the model is fine-tuned on the small-sample target domain dataset D_T^{small} ($D_T^{\text{small}} \ll D_T^{\text{large}}$):

$$\theta_T = \arg \min_{\theta \in \mathcal{F}(\theta_S)} \mathbb{E}_{(\mathbf{x}, y) \sim D_T^{\text{small}}} [\mathcal{L}_T(f(\mathbf{x}), y)] \quad (7)$$

where $\mathbb{E}_{(\mathbf{x}, y) \sim D}$ is the expectation of the loss function, $\mathcal{L}(\cdot)$ over the dataset D , θ_S is the source domain parameters, θ_T

the target domain parameters, and $\mathcal{F}(\theta_S)$ the function space of models initialized with the source domain parameters θ_S .

Transfer learning leverages the knowledge acquired from the source domain to improve the generalization ability of the model under data-deficient scenarios in the target domain.

III. PROPOSED METHOD

A. The Framework of the Proposed Method

The framework of the proposed method is shown in Fig. 1.

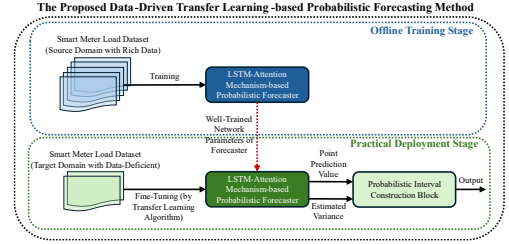


Fig. 1. The framework of the proposed method.

In Fig. 1, the proposed method builds a well-trained LSTM-attention mechanism-based probabilistic forecaster using a source domain dataset with rich smart meter load data during the offline training stage. In the offline training stage, the LSTM-attention mechanism-based probabilistic forecaster learns a wealth of knowledge about smart meter load forecasting, enabling it to quickly adapt and achieve excellent performance when faced with new forecasting tasks in data-deficient scenarios. In addition, a large amount of related task learning makes the LSTM-attention mechanism-based probabilistic forecaster more robust than traditionally trained load forecaster.

After being well-trained on the source domain data, the LSTM-attention mechanism-based probabilistic forecaster is then fine-tuned using a transfer learning algorithm on the target domain smart meter data during the practical deployment stage. This process effectively transfers knowledge from data-rich domains to data-deficient domains. Finally, the point prediction value and the corresponding estimated error variance generated by the forecaster are fed into the probabilistic interval construction module to generate the final probabilistic interval.

B. LSTM-Attention Mechanism-based Probabilistic Forecaster

In this section, the objective is to develop a probabilistic forecasting model with high accuracy. To achieve this, the LSTM-attention mechanism-based probabilistic forecaster is adopted. The LSTM module captures temporal dependencies in the data by extracting long-term trends and periodic patterns. The attention mechanism assigns appropriate weights to the LSTM output sequence, enabling the model to focus on critical time steps and thereby improve predictive accuracy. This hybrid approach allows the model to better capture temporal relationships and allocate attention weights globally and effectively, enhancing the reliability of the probabilistic forecasts.

The architecture of the LSTM-attention mechanism-based probabilistic forecaster consists of four main components, as shown in Figure 2.

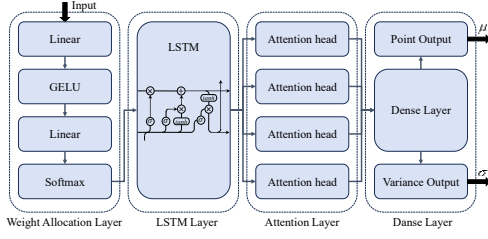


Fig. 2. Structure of the LSTM-attention mechanism-based probabilistic forecaster.

The first component is the weight allocation layer, which assigns weights to input features through a latent projection and the Softmax function. This layer is responsible for identifying the most informative features for prediction. The feature weighting is expressed as follows:

$$\mathbf{w} = \text{Softmax}(\text{Linear}(\mathbf{x})) \quad (8)$$

where $\mathbf{x}$ is the input vector, $\mathbf{w}$ is the corresponding feature weights matrix, $\text{Linear}(\cdot)$ represents the linear layer transformation applied to the input features, and $\text{Softmax}(\cdot)$ is the Softmax function that normalizes these raw weights.

Subsequently, the LSTM layer models the temporal dependencies of the input sequence by processing it step by step and updating its hidden state accordingly.

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad (9)$$

where x_t is the input at time t , h_{t-1} is the previous hidden state, h_t is the current hidden state, and $\text{LSTM}(\cdot)$ represents the LSTM transformation applied to the input features.

Following this, the attention layer, specifically a four-head attention mechanism, reweights the outputs of the LSTM layer. By calculating energy scores and normalizing them, the attention layer emphasizes critical time steps, as expressed in the equation:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \quad (10)$$

where e_t is the energy score, $\exp(e_t)$ is the exponential of the energy score e_t , and α_t is the attention weight for time t .

Finally, the dense layer maps the attention-weighted output to the desired prediction space. As this is a probabilistic forecaster, it outputs both the point prediction value $\hat{y}$ and corresponding estimated error variance σ^2 of a Gaussian distribution:

$$\hat{y}, \sigma^2 = \text{Dense}(\mathbf{h}_T) \quad (11)$$

where $\mathbf{h}_T$ is the output matrix after attention processing, and $\text{Dense}(\cdot)$ represents the application of a dense layer.

During training, the LSTM-attention mechanism-based probabilistic forecaster is optimized using the NLL loss. This loss ensures the model not only fits the data but also quantifies its confidence. The advantage of using the NLL loss is its ability to reduce the penalization of large errors when the predicted variance is substantial, making the model more robust to outliers or volatile data. The NLL loss $\mathcal{L}_{\text{NLL}}$ is given by:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\log \sigma^2 + \frac{(y - \hat{y})^2}{\sigma^2} \right) \quad (12)$$

where y is the true value.

The LSTM-attention mechanism-based probabilistic forecaster is expected to accurately predict user electricity load and provide credible uncertainty estimates, such as a 95% confidence interval. This probabilistic approach aids decision-makers in better assessing the reliability of forecasts and making informed decisions accordingly. Simulation results indicate that the LSTM-attention mechanism-based probabilistic forecaster significantly outperforms baseline models in both point prediction accuracy and variance estimation, thereby achieving high-precision probabilistic forecasting.

C. Transfer Learning Algorithm for Data-Deficient Scenarios

In data-deficient scenarios, directly training models from scratch often leads to reduced prediction accuracy and slow convergence. To address these challenges, the Fine-tune algorithm, which is shown in Figure 3, is adopted as the transfer learning algorithm for data-deficient load forecasting. This algorithm demonstrates three distinct advantages: computational efficiency, architectural lightweight design, and adaptability across diverse data regimes.

The core concept involves initializing the target domain model with parameters derived from the source domain, followed by progressive training using target domain data to capture domain-specific scale characteristics and temporal dependencies. In the target domain, the fine-tuning algorithm only updates the parameters of some layers. Let θ represent the model parameters, where $\theta = \{\theta_f, \theta_t\}$, θ_f are the frozen parameters, and θ_t are the parameters that need to be fine-tuned.

The objective of fine-tuning is to minimize the negative log-likelihood loss function on the target domain D_T :

$$\theta_{t*} = \arg \min_{\theta_t} \mathcal{L}_{\text{NLL}}(y_t, p(y_t | \mathbf{x}_t; \theta_f, \theta_t); D_T) \quad (13)$$

where θ_{t*} is the optimal fine-tuned parameters matrix, $\mathbf{x}$ is the input features matrix, y is the future load, and $\mathcal{L}_{\text{NLL}}(\cdot)$ is the predicted probabilities for target domain inputs.

Finally, the prediction interval is constructed using a probabilistic interval construction block defined by the standard normal quantile z :

$$[\hat{y}_{\text{lower}}, \hat{y}_{\text{upper}}] = [\hat{y} - z\sigma, \hat{y} + z\sigma] \quad (14)$$

where $z = 1.96$ corresponds to a 95% confidence level.

The transfer learning algorithm for data-deficient load forecasting is expected to enhance training efficiency and prediction performance while also demonstrating strong generalization capability. Simulation results indicate that it achieves higher predictive accuracy and faster training convergence under data-deficient scenarios, compared to direct training and other transfer learning algorithms in multiple experiments.

IV. RESULTS

A. Dataset and Evaluation Metrics

This study utilizes smart meter electricity consumption data obtained from residential users in France, which were collected by the distribution system operator Enedis over the period spanning from 2011 to 2014.

To evaluate the performance of the proposed method, this paper uses the following metrics: continuous ranked probability score (CRPS), prediction interval coverage probability (PICP), average coverage error (ACE), mean prediction interval width (MPIW), and prediction interval normalized average width (PINAW). The formulas are as follows:

$$CRPS(F, y) = \int_{-\infty}^{\infty} (F(z) - 1\{y \leq z\})^2 dz \quad (15)$$

$$PICP = \frac{1}{N} \sum_{i=1}^N 1\{y_i^L \leq y_i \leq y_i^U\} \quad (16)$$

$$ACE = |PICP_\alpha - \alpha| \quad (17)$$

$$MPIW = \frac{1}{N} \sum_{i=1}^N (y_i^U - y_i^L) \quad (18)$$

$$PINAW = \frac{1}{N(\max(y) - \min(y))} \sum_{i=1}^N (y_i^U - y_i^L) \quad (19)$$

where N is the total number of samples, y is the actual observation value, y_i is the true value of the i -th sample, y_i^L and y_i^U are the lower and upper bounds of the prediction interval for the i -th sample, z is a random variable representing the predicted load, $\hat{F}(z)$ is the predicted CDF evaluated at z , $\max(y)$ and $\min(y)$ are the maximum and minimum values of y .

B. Probabilistic Forecasting

TABLE I. LOAD FORECASTING RESULT OF THE THREE MODELS

Model	CRPS	PICP	MPIW	PINAW	ACE
The Proposed	0.01177	0.9508	347.7	0.127	0.0008
QR based	0.01256	0.9595	380.3	0.139	0.0095
MC based	0.01259	0.8285	262.6	0.096	0.1215

Table 1 summarizes the probabilistic forecasting performance of three models: the proposed model, the QR-based model [19], and the MC-based model [20] across five evaluation metrics. The proposed model achieves the best overall results, with the lowest CRPS (0.01177), smallest ACE (0.0008), and narrowest intervals (PINAW of 0.127), while maintaining a high PICP of 0.9508, indicating strong accuracy and reliability. The QR model attains the highest PICP (0.9595), showing strong reliability but with wider, more conservative intervals. MC performs worst, with the lowest coverage (PICP 0.8285) and highest ACE, implying greater uncertainty despite relatively narrow intervals.

Figures 3(a) to (c) show the 95% prediction intervals and true values for all models. The proposed model consistently encloses true values with well-calibrated, stable intervals, whereas baseline models display higher variability and less reliable coverage, deviating more from the ground truth.

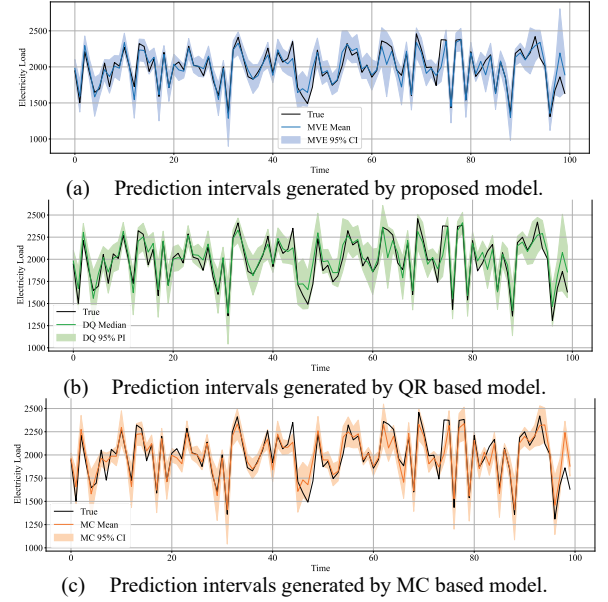


Fig. 3. Prediction intervals for electricity load forecasting.

In summary, the proposed model delivers the most accurate and sharp probabilistic forecasts, making it the most suitable model for this task.

C. Transfer Learning

Figure 4(a) to (d) present a comparative analysis of four target domain model training algorithms—Fine-tune, Direct Training, Feature Extractor, and MMD Transfer—under varying target domain sizes, with a focus on achieving efficient and reliable probabilistic load forecasting in low-data regimes.

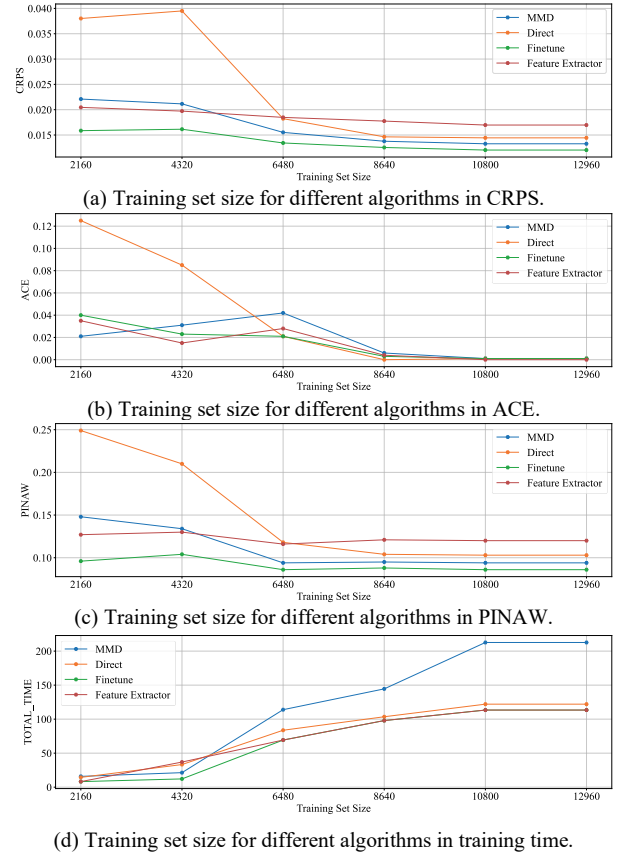


Fig. 4. Performance comparison of four algorithms in data-deficient scenario.

At a training size of 10800 samples, the Fine-tune method achieves a CRPS of 0.01203, PICP of 0.949, and ACE of 0.086, indicating superior predictive accuracy and well-calibrated uncertainty estimation. It also maintains a relatively narrow prediction interval width (MPIW: 1058.37) with moderate computational time (113.36 seconds).

By contrast, the Direct method records a higher CRPS (0.01444) and ACE (0.103). Feature and MMD methods perform between these two extremes but do not surpass the Fine-tune approach. Overall performance improves with increasing training data, although Fine-tune consistently exhibits notable advantages, particularly in data-scarce scenarios, demonstrating robustness and effectiveness.

In conclusion, the Fine-tune algorithm provides the best balance of accuracy, calibration, and efficiency for probabilistic load forecasting under limited datasets conditions.

Remark: While the Fine-tune algorithm outperforms the MMD algorithm in this context, the MMD algorithm can effectively align the distributions between the source and target domains, and it performs strongly when data are abundant. This alignment enables the model to generalize better across different datasets, thus holding great potential. However, its performance is limited in small-sample scenarios due to the risk of overfitting. In the context of this study, the MMD algorithm performs poorly, and future work may need to adopt strategies such as regularization or data augmentation to enhance its robustness in data-deficient scenarios.

V. CONCLUSION

This paper proposes a data-driven transfer learning-based probabilistic forecasting method combining LSTM and attention mechanisms to tackle prediction uncertainty quantification and insufficient target domain data. The method involves a two-stage training process: first, training the LSTM-attention mechanism-based probabilistic forecaster with abundant source domain data, followed by fine-tuning with limited target domain data for adaptation. The prediction interval is constructed by integrating point predictions with estimated variances. Simulation results using a public smart meter dataset demonstrate the method's effectiveness, showing superior performance in probabilistic load forecasting with lower CRPS, smaller ACE, narrower intervals, and higher PICP compared to benchmarks. The fine-tuning algorithm enhances its efficiency and robustness in data-scarce scenarios, making it a viable solution for practical probabilistic load forecasting.

REFERENCES

- [1] H. Hong and S. Fan, "Probabilistic electric load forecasting: A tutorial review," *International Journal of Forecasting*, vol. 32, no. 3, pp. 914–938, Jul. 2016.
- [2] S. Taieb, R. Hyndman, X. Wang, and H. Yang, "Forecasting uncertainty in electricity demand: Methods and applications," *European Journal of Operational Research*, vol. 291, no. 1, pp. 379–389, Mar. 2021.
- [3] A. Humeau, M. E. H. Benbouzid, and M. Boucherit, "Comparison of deep learning models for residential load forecasting," *Applied Energy*, vol. 276, p. 115436, Aug. 2020.
- [4] Y. Jiang, C. Wang, J. Liu, et al., "Transfer learning for short-term electric load forecasting under data insufficiency," *IEEE Access*, vol. 9, pp. 135365–135375, 2021.
- [5] D. C. Park, M. A. El-Sharkawi, R. J. Marks II, L. E. Atlas, and M. J. Damborg, "Electric load forecasting using an artificial neural network," *IEEE Trans. Power Syst.*, vol. 6, no. 2, pp. 442–449, May 1991.
- [6] B.-J. Chen, M.-W. Chang, and C.-J. Lin, "Load forecasting using support vector machines: A study on EUNITE competition 2001," *IEEE Trans. Power Syst.*, vol. 19, no. 4, pp. 1821–1828, Nov. 2004.
- [7] W. Kong et al., "Short-term residential load forecasting based on LSTM recurrent neural network," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 841–851, Jan. 2019.
- [8] K. Chen et al., "Short-term load forecasting with deep residual networks," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3943–3952, Jul. 2019.
- [9] S. Perna et al., "A Comparative Analysis of Empirical Copula and Quantile Regression Methods for Probabilistic Load Forecasting," in *2024 IEEE 18th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*, 2024, pp. 1–6.
- [10] D. Gan et al., "Embedding based quantile regression neural network for probabilistic load forecasting," *Journal of Modern Power Systems and Clean Energy*, vol. 6, no. 2, pp. 244–254, Mar. 2018.
- [11] Z. Ye et al., "Evaluation Study of Deep Learning-based Models on Probabilistic Power Load Forecasting," in *2023 IEEE 7th Conference on Energy Internet and Energy System Integration*, 2023, pp. 4312–4317.
- [12] Z. Zhang et al., "Long Short-Term Memory-Based Twin Support Vector Regression for Probabilistic Load Forecasting," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 1, pp. 1764–1778, Jan. 2025.
- [13] E. Lee and W. Rhee, "Individualized Short-Term Electric Load Forecasting With Deep Neural Network Based Transfer Learning and Meta Learning," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 3, pp. 1–10, May 2020.
- [14] Y. Fu et al., "On the Benefits of Transfer Learning and Reinforcement Learning for Electric Short-term Load Forecasting," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 4, pp. 1–10, July 2021.
- [15] Y. Tian et al., "Similarity-Based Chained Transfer Learning for Energy Forecasting," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 2, pp. 1–10, March 2019.
- [16] L. Cai et al., "Two-Layer Transfer-Learning-Based Architecture for Short-Term Load Forecasting," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 3, pp. 1–10, May 2020.
- [17] D. Wu and W. Lin, "Efficient Residential Electric Load Forecasting via Transfer Learning and Graph Neural Networks," *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2423–2431, May 2023.
- [18] W. Kong, Z. Y. Dong, D. J. Hill, F. Luo, and Y. Xu, "Short-term residential load forecasting based on LSTM networks," *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 841–851, Jan. 2019.
- [19] H. -Y. Lo, Y. -K. Wu, Q. -T. Phan and W. -S. Tan, "A Novel QR-based Probabilistic Forecasting Method for Solar Power Generation," in *IEEE Transactions on Industry Applications* (2025).
- [20] S. Sato, M. Takanashi, K. Indo, N. Nishihara, H. Ichikawa and H. Watanabe, "Two-Stage Probabilistic Short-Term Wind Power Prediction Using Neural Network with MC Dropout and Control Information," *PCIM Europe digital days 2021; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, Online, 2021, pp. 1–8